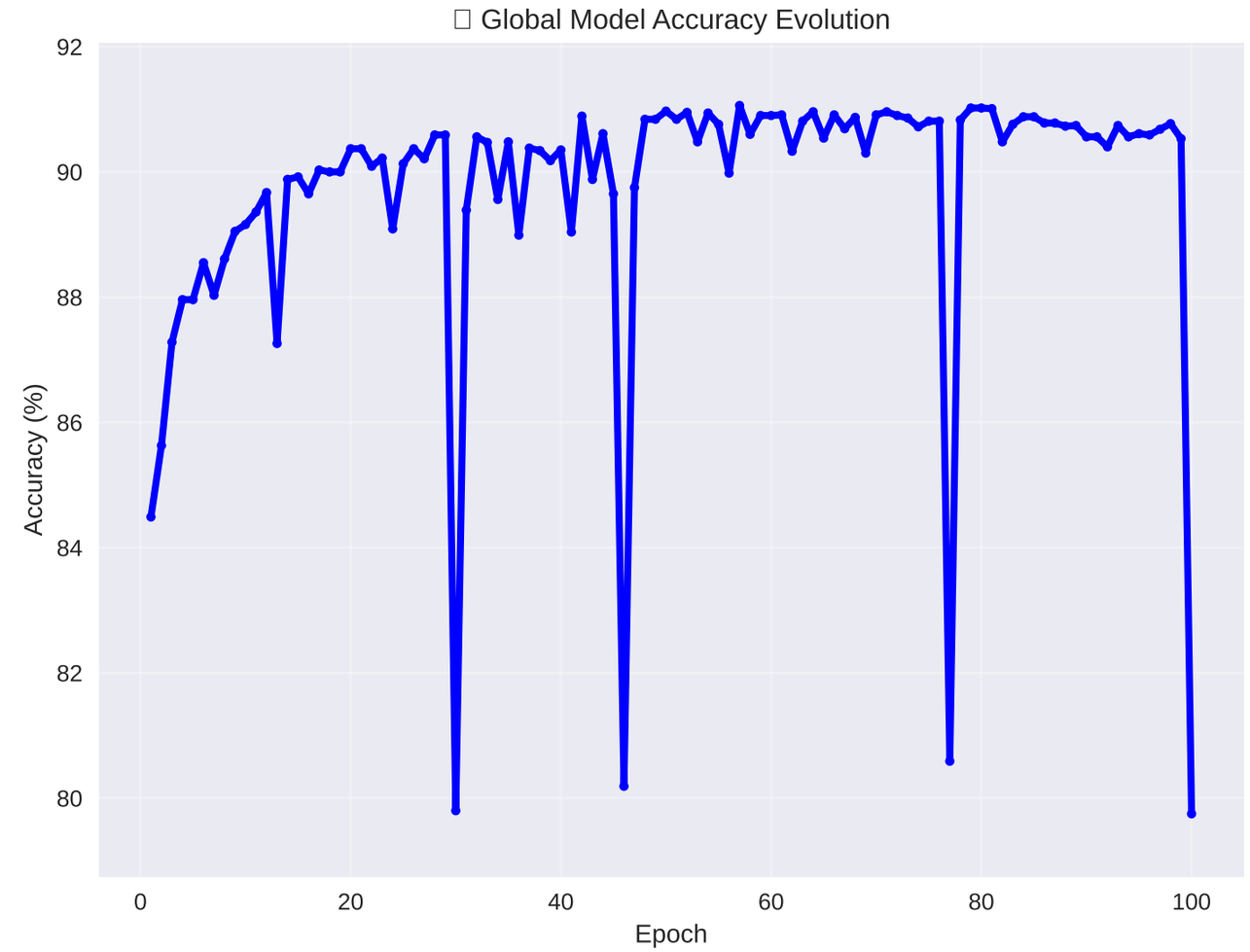
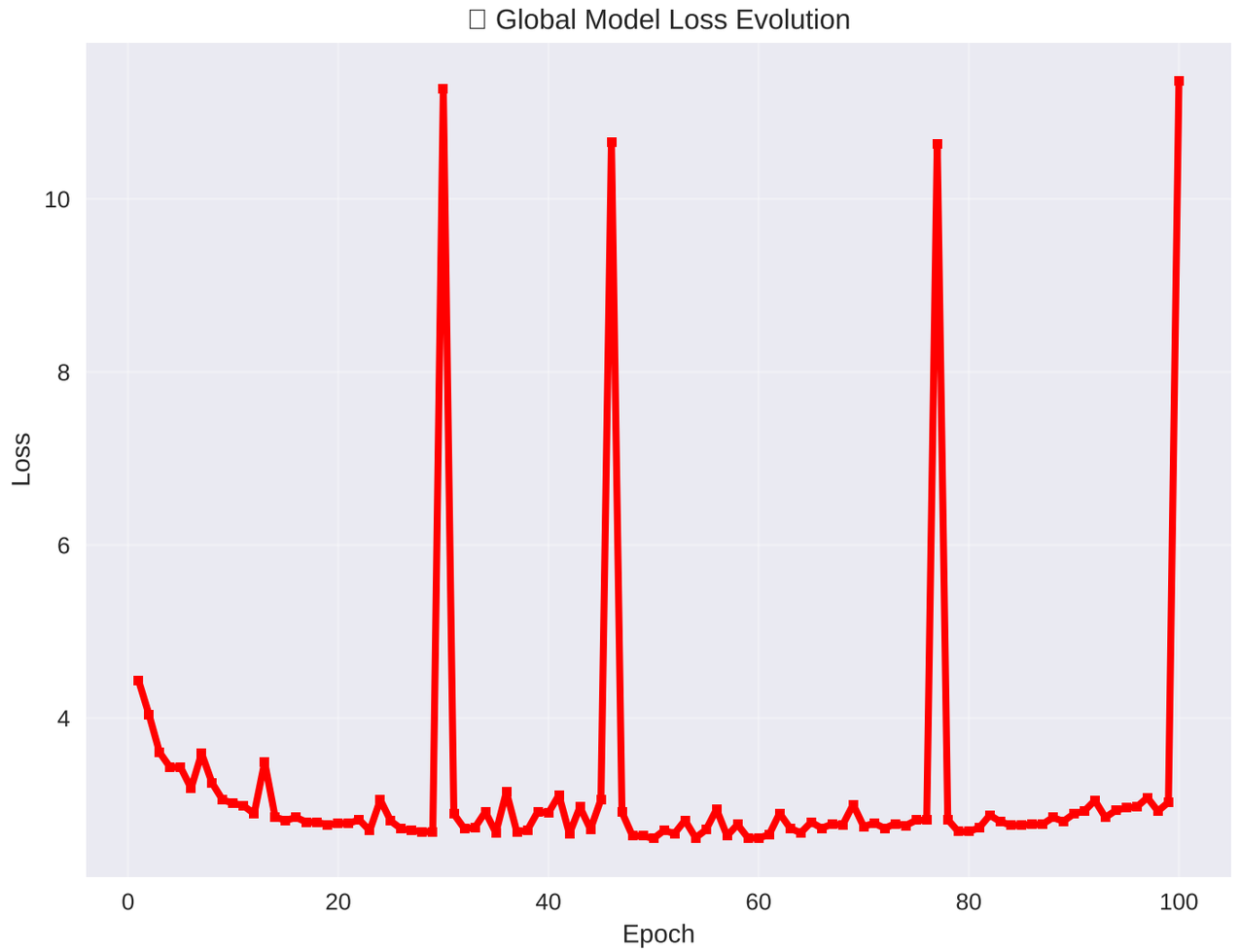
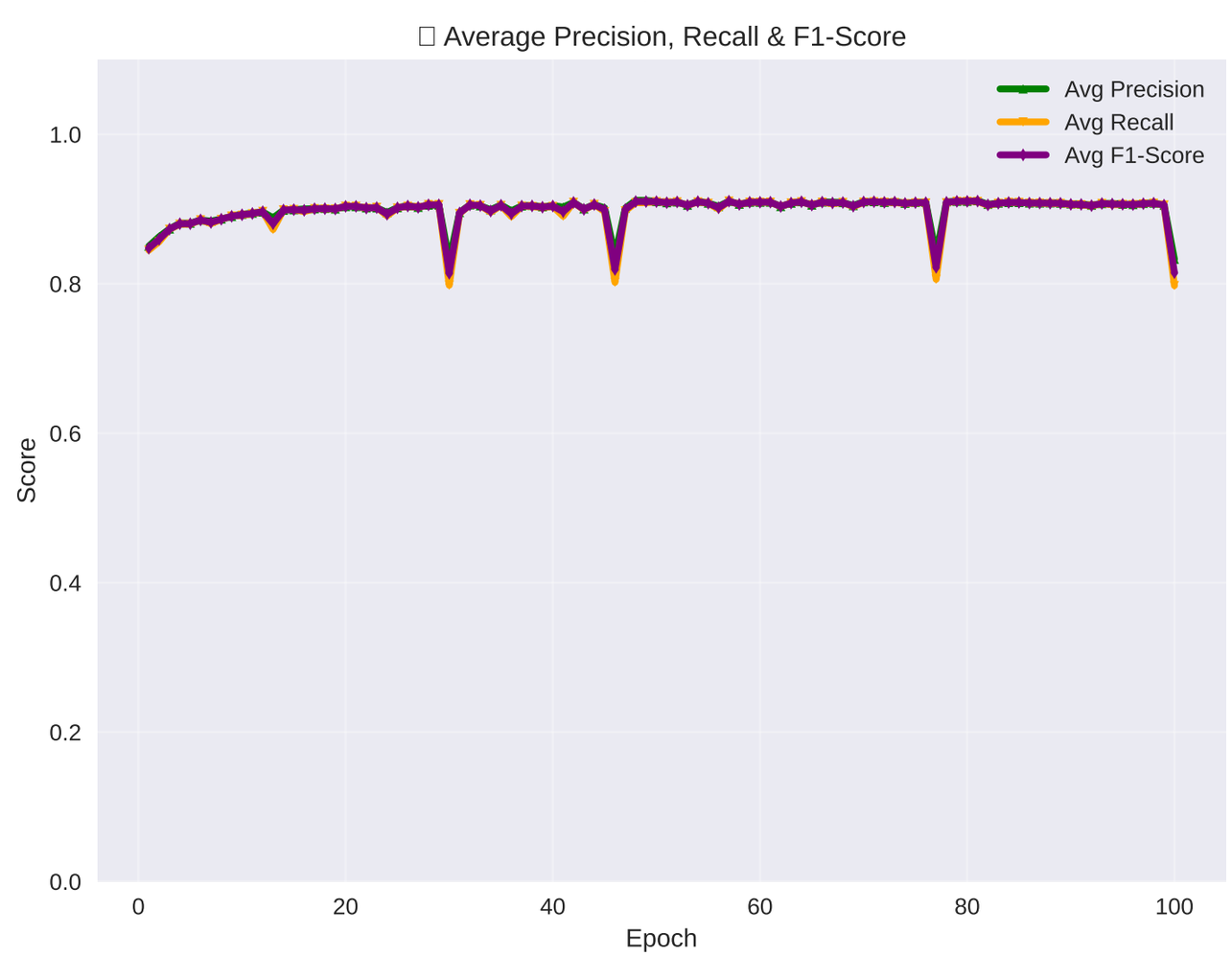
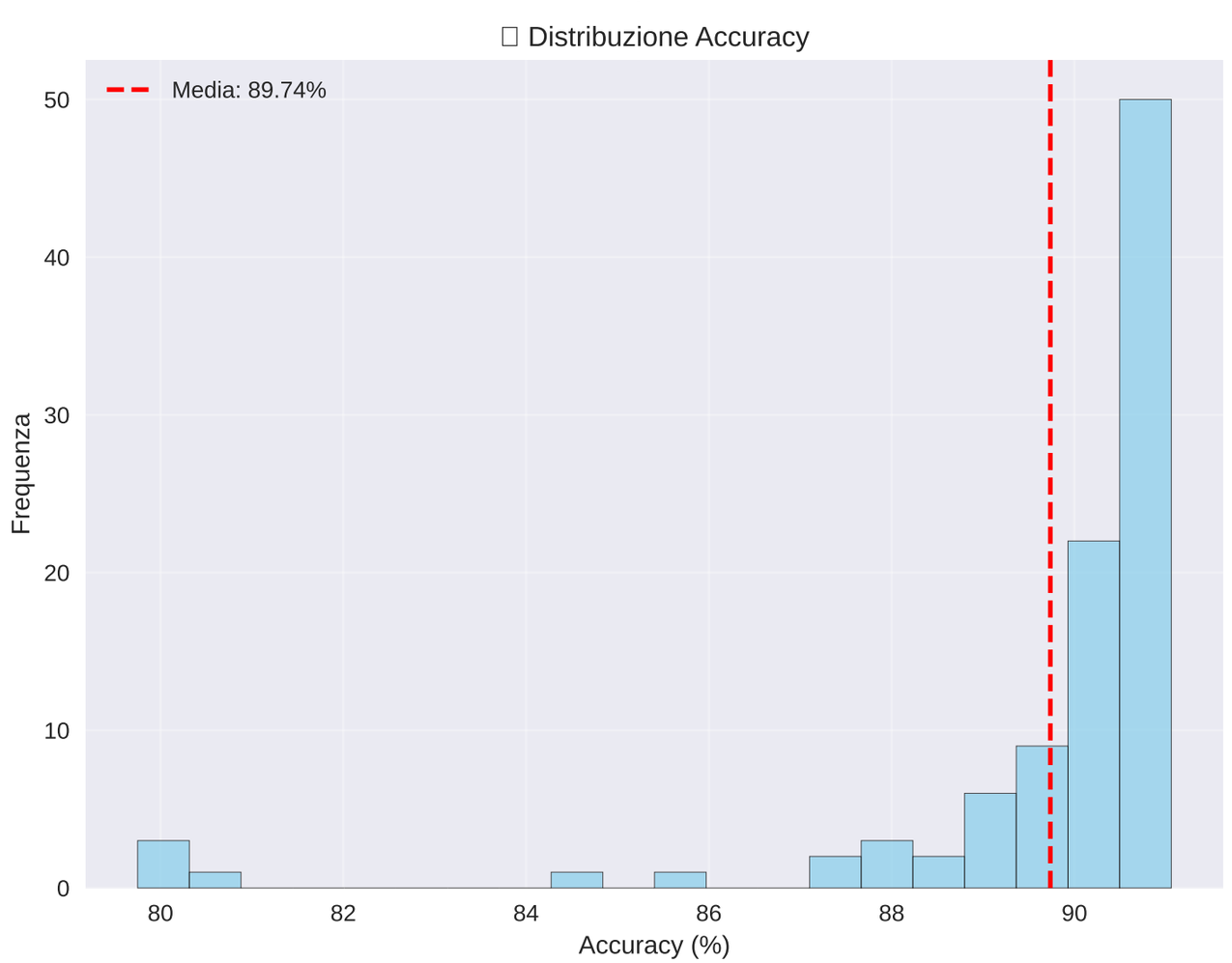
Federated Learning - Main Metrics Overview







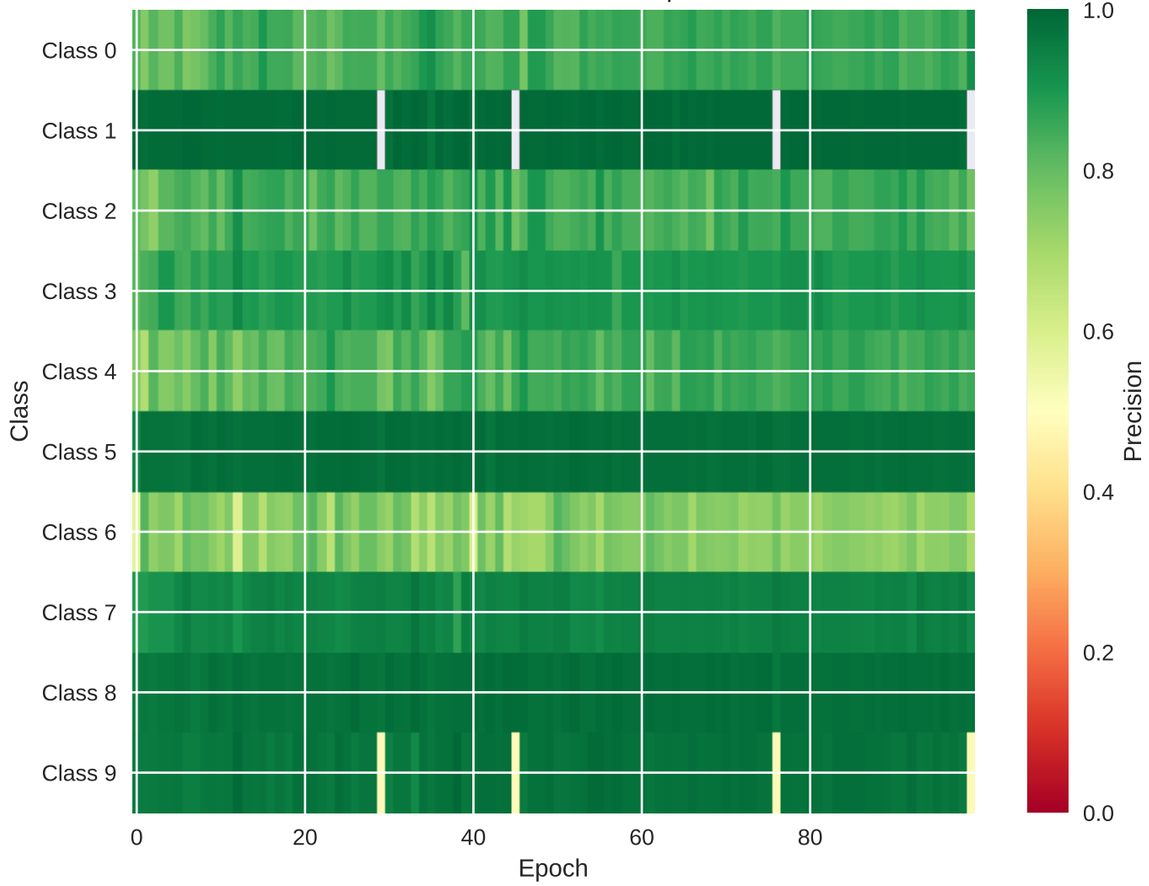


Per-Class Precision Analysis

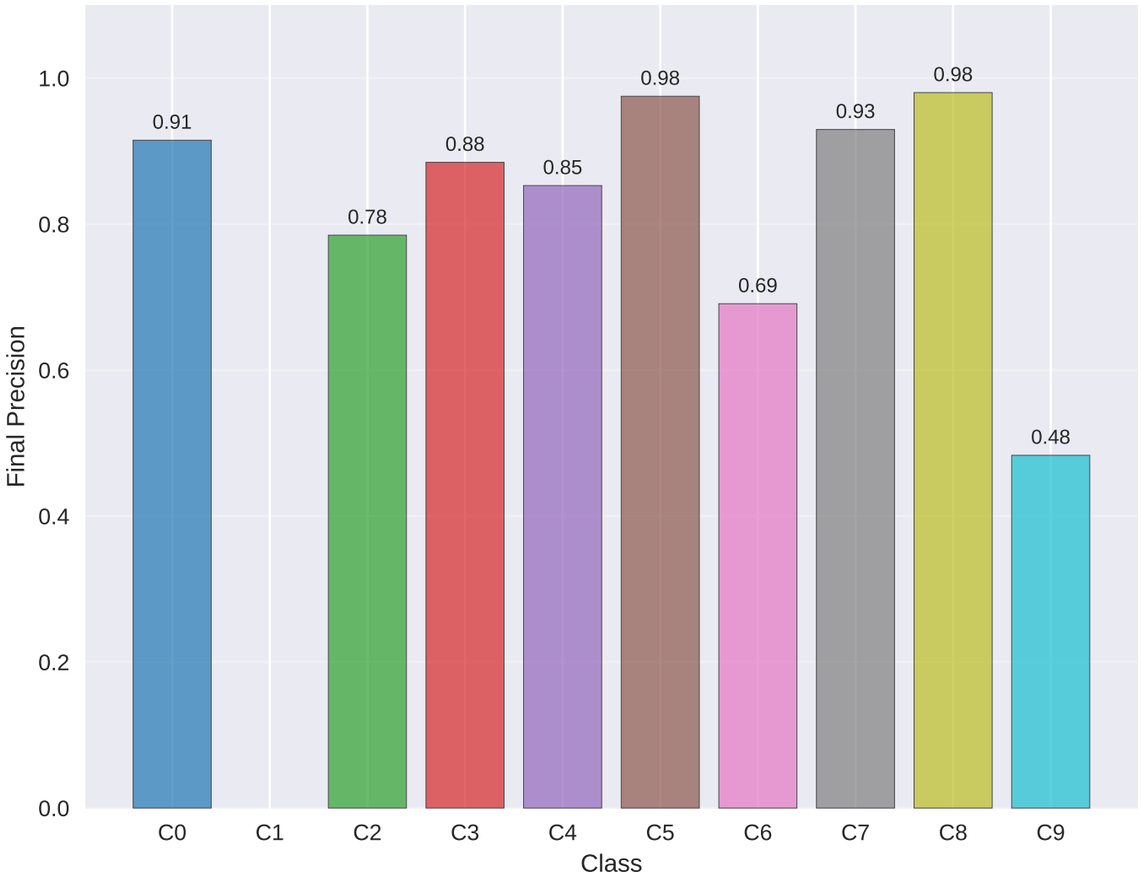
□ Per-Class Precision Evolution



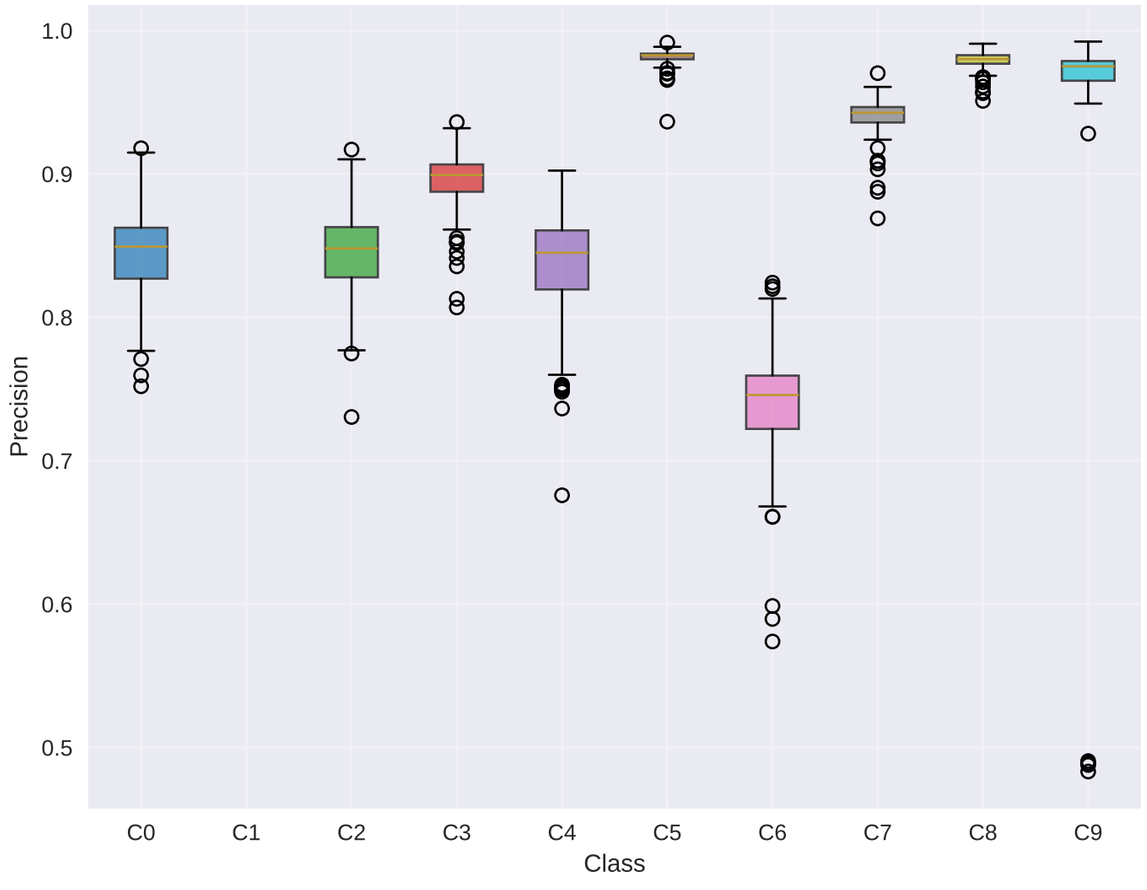
□ Precision Heatmap



□ Final Precision per Class

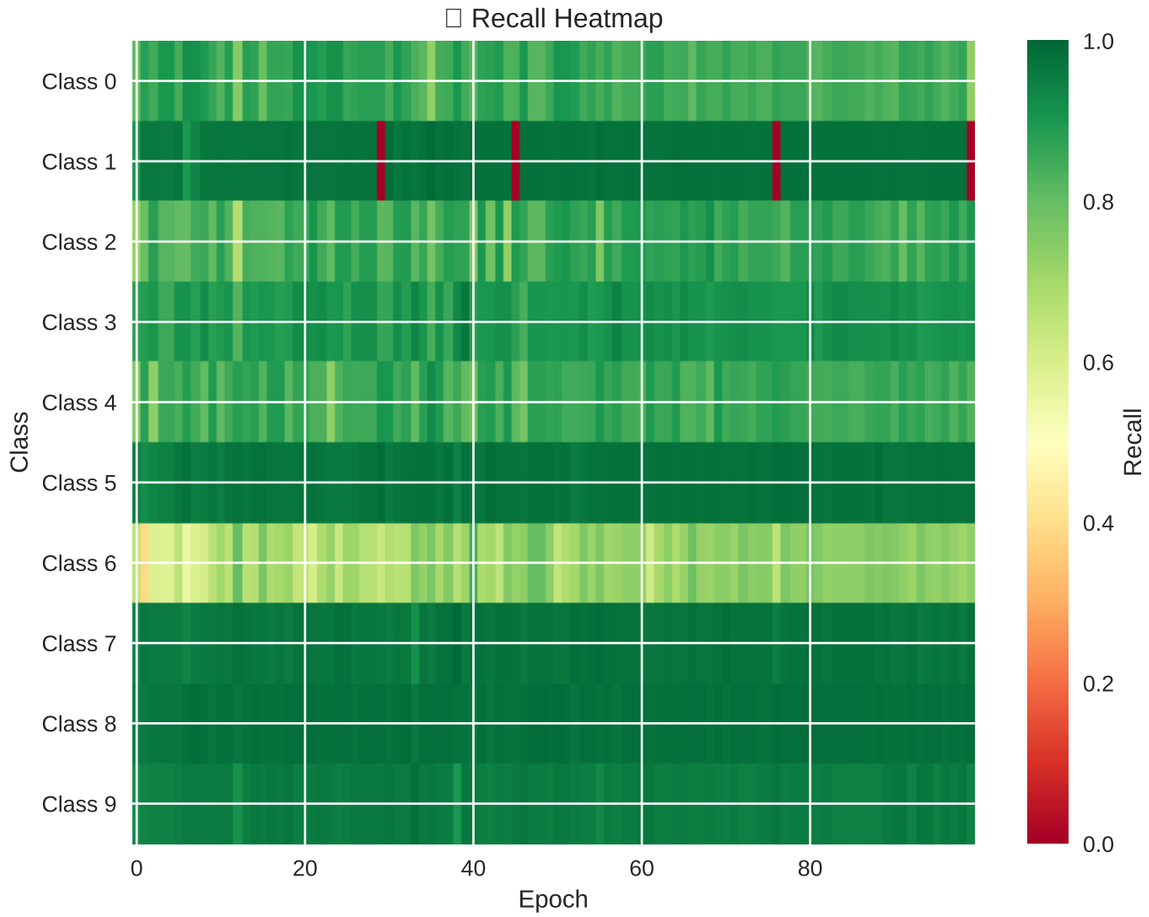


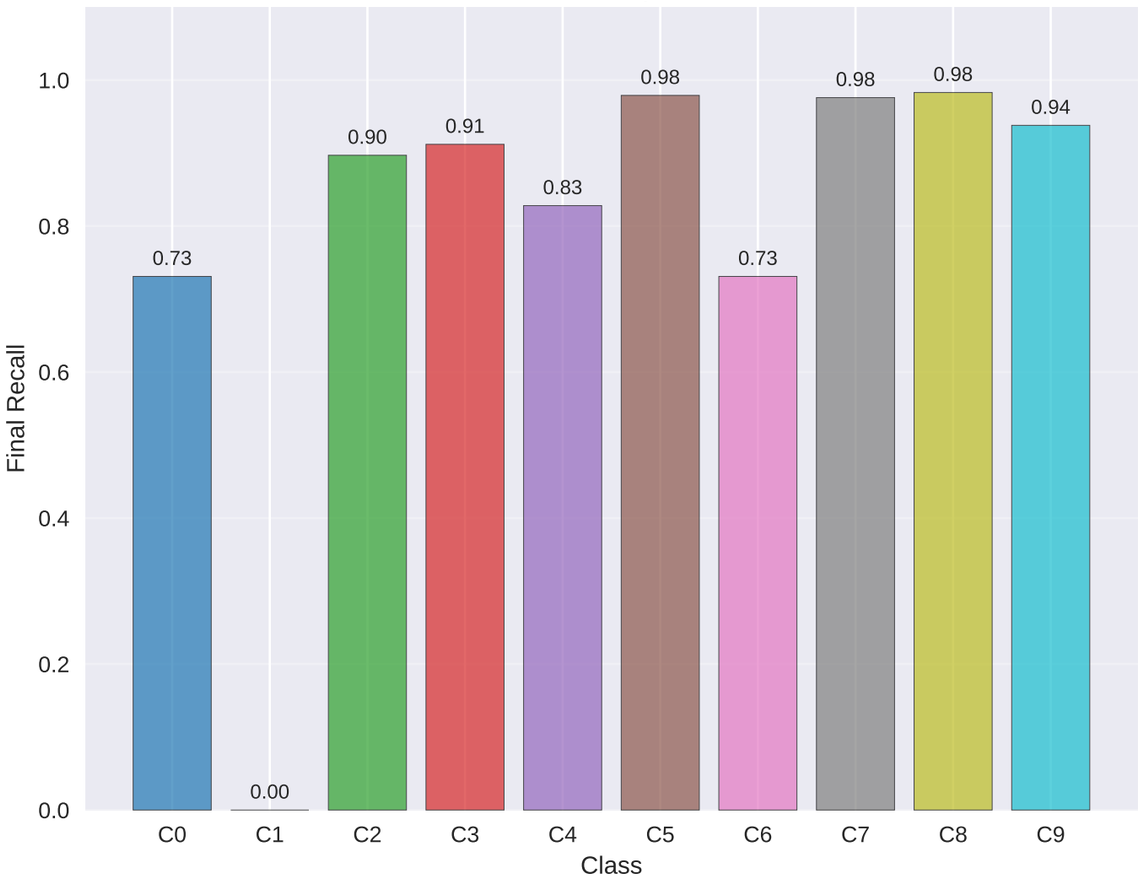
□ Precision Distribution per Class

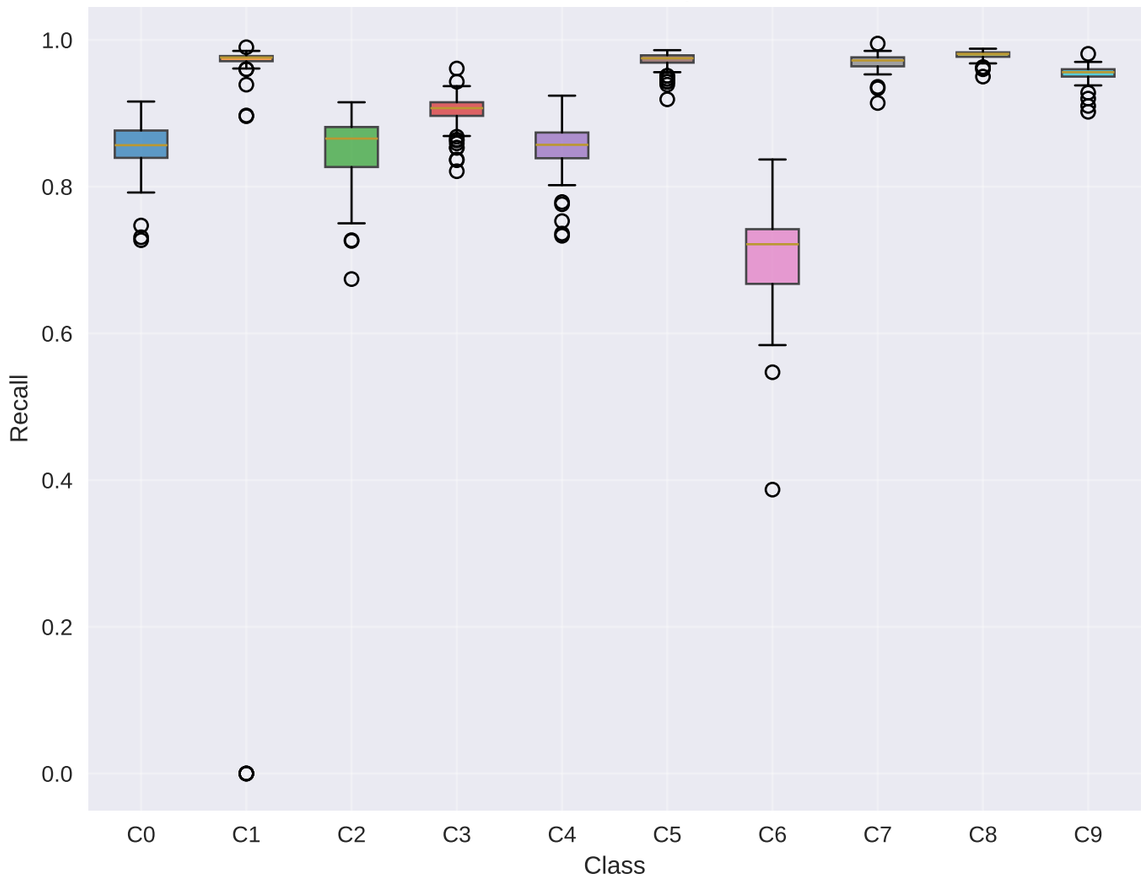


Per-Class Recall Analysis



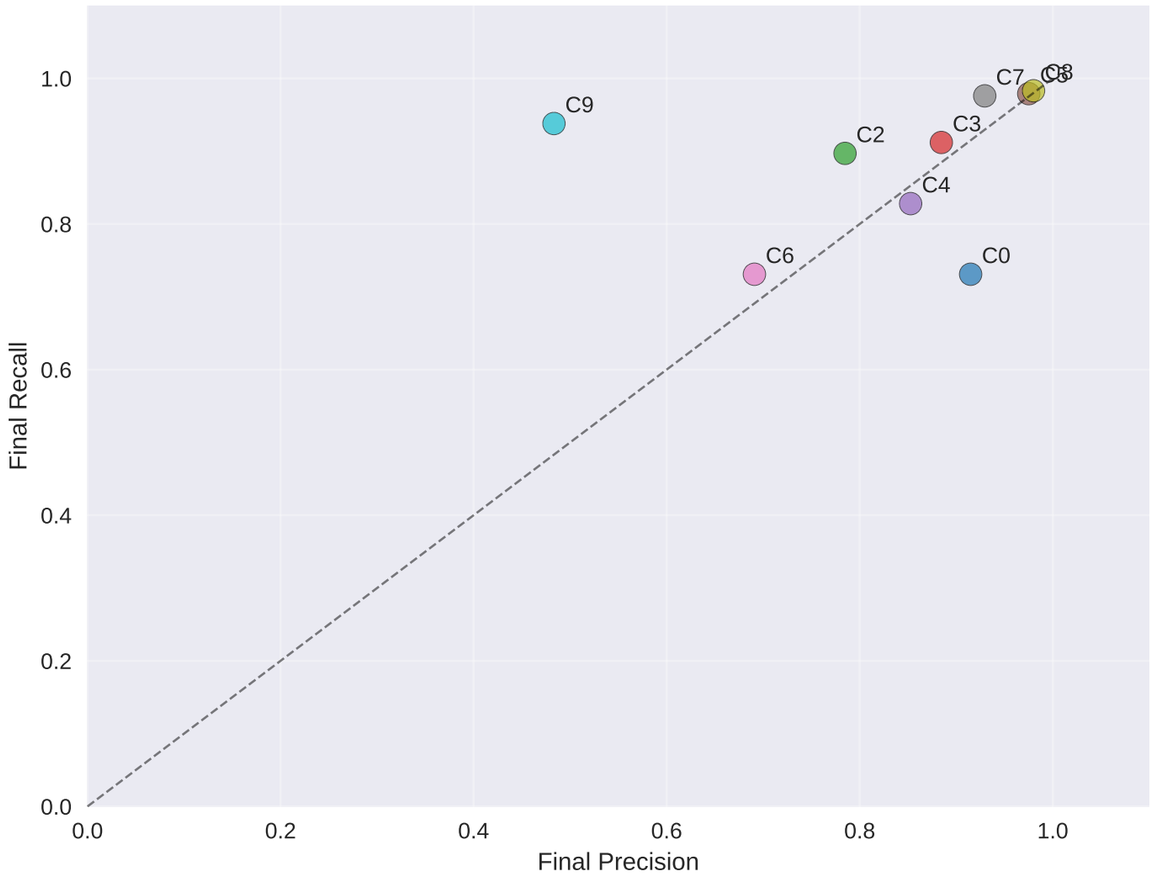




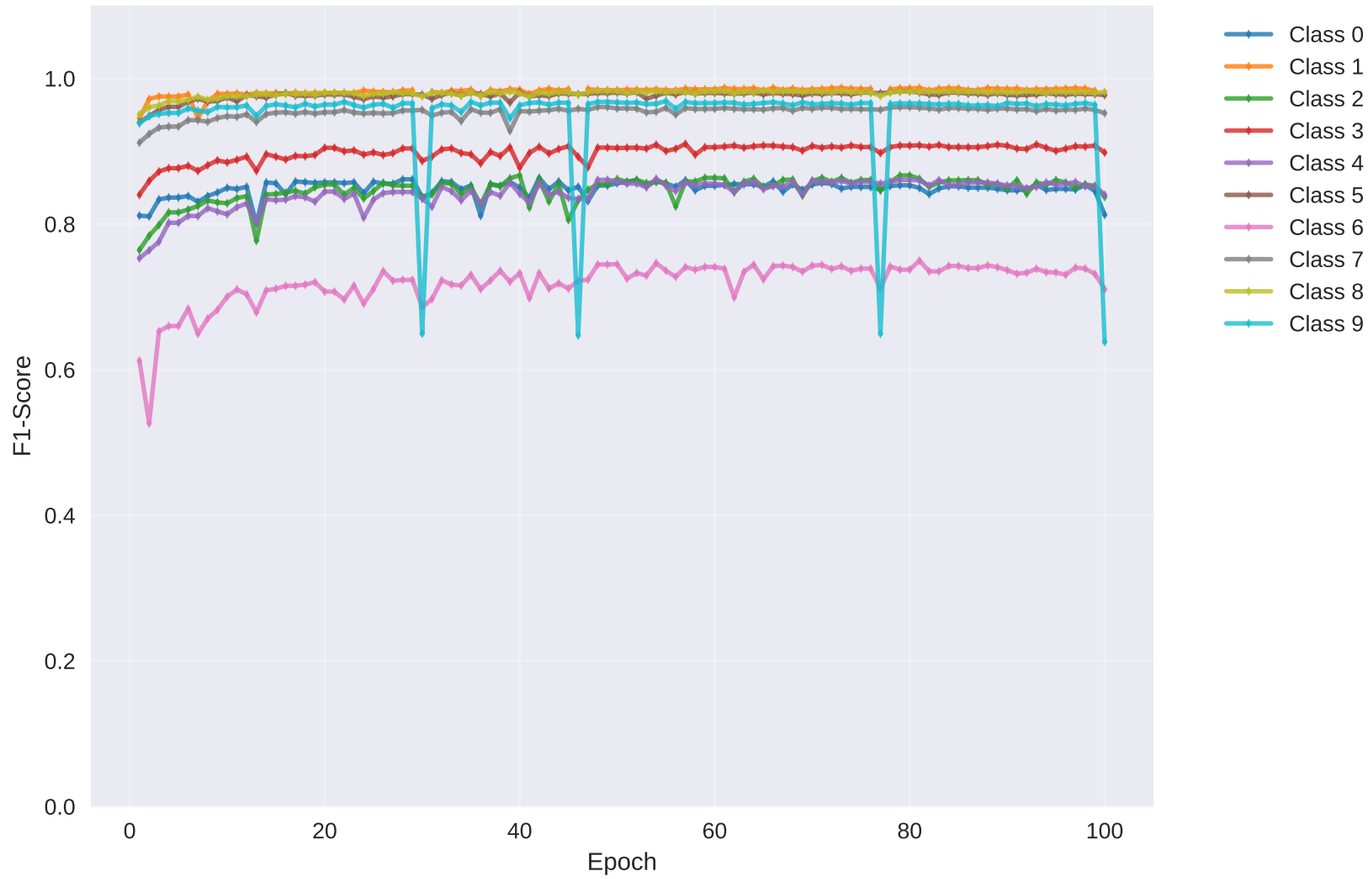


Comparative Analysis

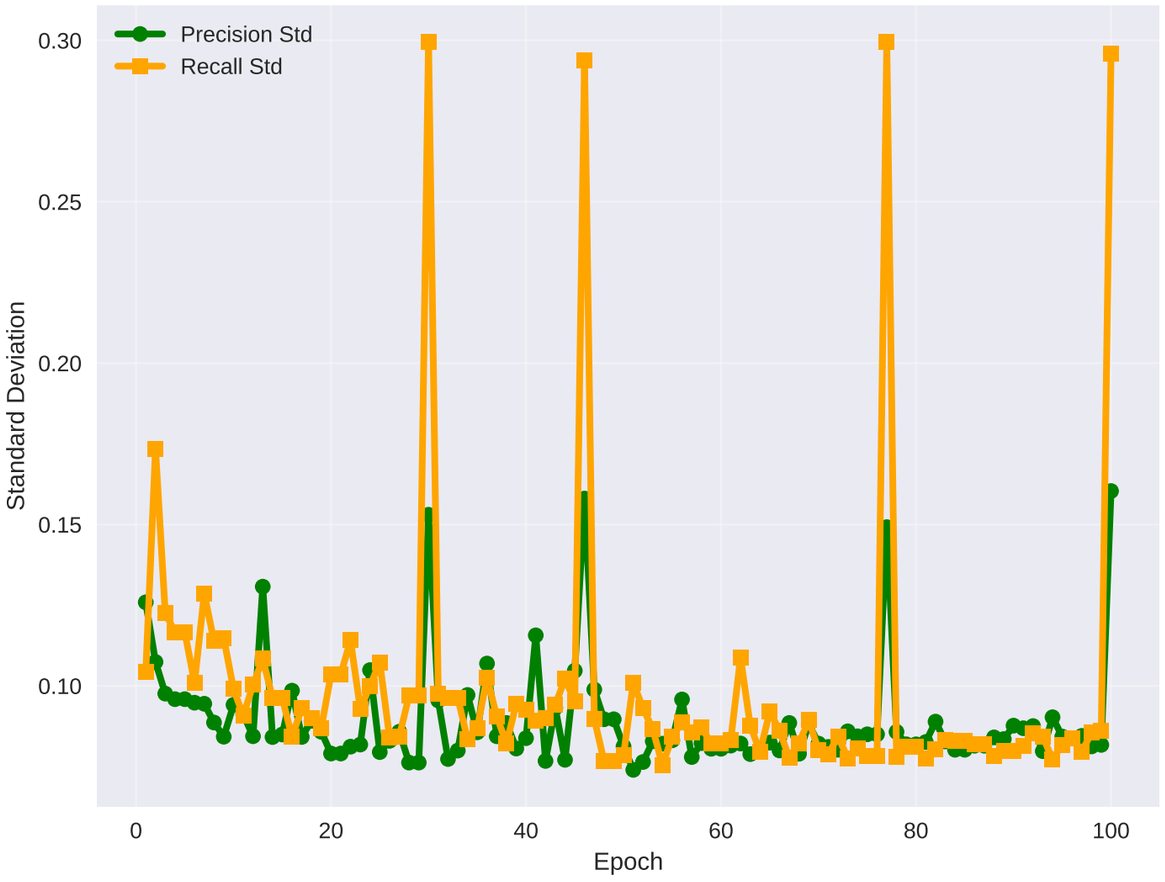
□ Precision vs Recall (Final Values)



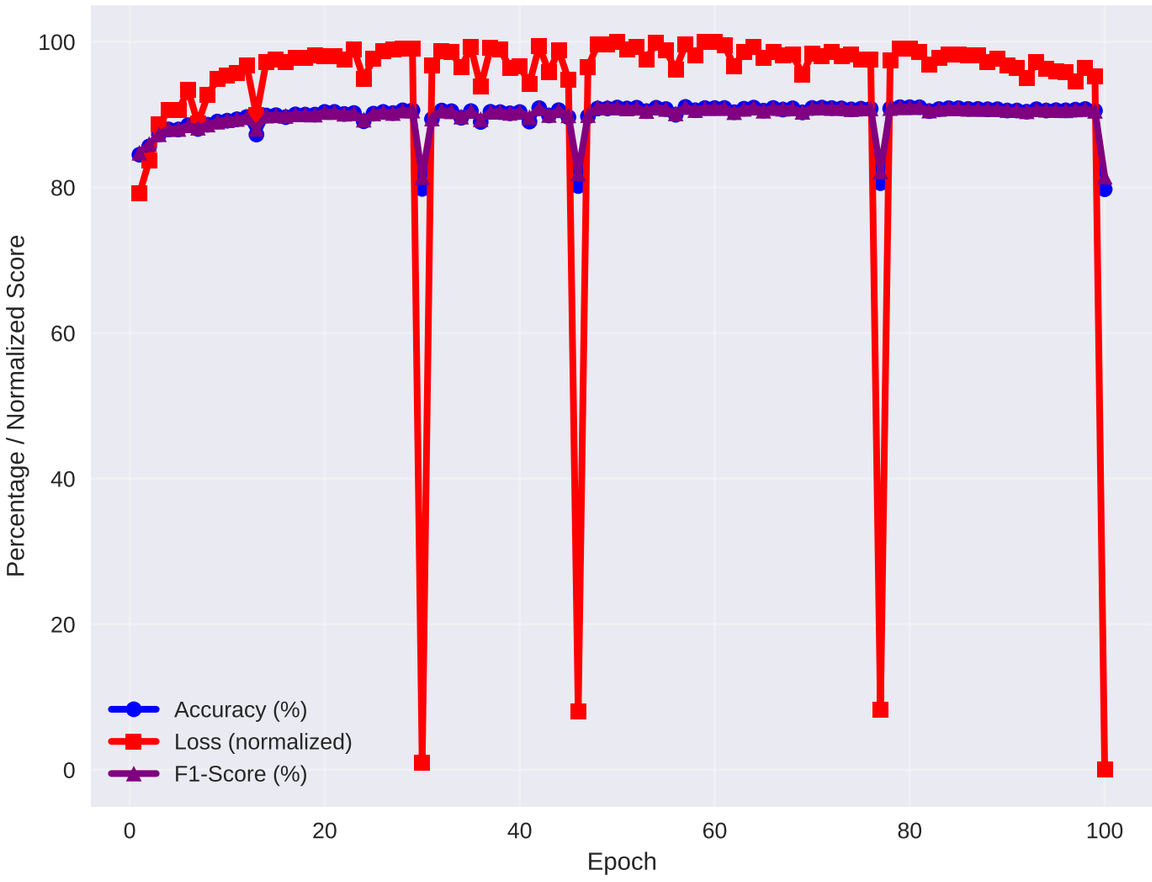
□ Per-Class F1-Score Evolution



□ Inter-Class Variance Evolution



□ Learning Curves Comparison



Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-05 19:30:16
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 79.75%
• Loss: 11.3664
• Average Precision: 0.833
• Average Recall: 0.797
• Average F1-Score: 0.815

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: -4.74%
• Riduzione Loss: -6.9330
• Stabilità Accuracy (CV): 0.0252

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (nan)
• Peggiore Precision: Classe 1 (nan)
• Migliore Recall: Classe 8 (0.983)
• Peggiore Recall: Classe 1 (0.000)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.915, R=0.731, F1=0.813
• Classe 1: P=nan, R=0.000, F1=nan
• Classe 2: P=0.785, R=0.897, F1=0.837
• Classe 3: P=0.885, R=0.912, F1=0.898
• Classe 4: P=0.853, R=0.828, F1=0.840
• Classe 5: P=0.975, R=0.979, F1=0.977
• Classe 6: P=0.691, R=0.731, F1=0.710
• Classe 7: P=0.930, R=0.976, F1=0.952
• Classe 8: P=0.980, R=0.983, F1=0.982
• Classe 9: P=0.483, R=0.938, F1=0.638

VALUTAZIONE QUALITATIVA:
• Performance Globale: DA MIGLIORARE
• Stabilità Training: MODERATAMENTE STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
• Considerare aumento epoche o tuning iperparametri
• Implementare class balancing o weighted loss
```